

C Programming Curriculum



COURSE CURRICULUM

C Programming Curriculum

Goal: Learn basic C programming concepts with fun examples and mini projects, enabling beginners to build small programs by themselves.

Week 1 – Getting Started with C

Day 1 – Introduction to Programming & C

- What is programming?
- What is C language?
- How to set up a C programming environment (like Code::Blocks or online compiler).
- Write your first program:

```
#include <stdio.h>
```

```
int main() {  
    printf("Hello, World!\n");  
    return 0;  
}
```

Day 2 – Basics of C Syntax

- #include, main() function, return 0
- Comments in C
- Printing text using printf
- Taking input using scanf

Day 3 – Variables & Data Types

- int, float, char
- Declaring variables
- Example: Store and print your age, height, and initial

Day 4 – Basic Arithmetic Operations

- +, -, *, /, %
- Simple calculator program

COURSE CURRICULUM

Day 5 – Simple Project: Interactive Calculator

- Let user input two numbers and select operation (+, -, *, /)

Week 2 – Logic Building

Day 6 – Conditional Statements (if / else)

- if, else if, else
- Example: Check if a number is positive, negative, or zero

Day 7 – Comparison Operators

- ==, !=, >, <, >=, <=
- Example: Find the largest of 3 numbers

Day 8 – Logical Operators

- &&, ||, !
- Example: Check if a number is between 1 and 100

Day 9 – Simple Project: Number Guessing Game

- Program generates a random number (1 to 10)
- User tries to guess it
- Give hints (too high/too low)

Day 10 – Looping Basics (for / while)

- for loop: Counting from 1 to 10
- while loop: Repeat until condition is false

COURSE CURRICULUM

Week 3 – Functions & Arrays

Day 11 – Functions in C

- What is a function and why use it
- Writing a simple function (e.g., a function that adds two numbers)

Day 12 – Arrays – Introduction

- Declare an array
- Store multiple numbers
- Example: Store 5 test scores and print them

Day 13 – Simple Array Operations

- Sum of all elements
- Find maximum value in the array

Day 14 – Simple Project: Student Score Program

- Let user input scores of 5 students
- Display the average score and highest score

Day 15 – Revision + Mini Quiz

- Multiple choice quiz on concepts learned
- Short coding exercises

Week 4 – Fun with C

Day 16 – Introduction to Strings

- What is a string in C
- `char name[50];`
- `printf` vs `scanf` with strings

COURSE CURRICULUM

Day 17 – String Operations

- `strlen()`, `strcpy()`, `strcmp()`
- Example: Check if entered password matches preset password

Day 18 – Simple Project: Password Checker

- Ask user to enter a password
- Compare with preset password
- Print "Access Granted" or "Access Denied"

Day 19 – Pointers (Basic Concept)

- Simple explanation of pointers using analogy (pointer is an address)
- Example: Swap two numbers using pointers

Day 20 – Final Project: Simple Mini-Application

- Build a small menu-driven app
- Example:
 1. Add Student Score
 2. View Scores
 3. Exit

Optional Extensions (Bonus)

- File handling (Store data to file and read back)
- Simple games: Rock-Paper-Scissors
- Build a small text-based quiz game

Final Outcome

By the end of this curriculum, kids will be able to:

- Write simple programs in C
- Use variables, conditions, loops, arrays, and functions
- Build small interactive applications
- Understand basics of pointers and string operations



Scan the QR to know more